

DATA ANALYTICS INTERNSHIP PORTFOLIO PROJECT

# HR Analytics Pipeline

Employee Attrition & Workforce Analysis

*MySQL to Python: an end to end HR data analytics workflow*

915

EMPLOYEE RECORDS

8

LINKED DATA TABLES

15.96%

OVERALL ATTRITION

5

ANALYTICS CHARTS

# What This Deck Covers

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# Introduction & Objective

A complete HR analytics pipeline built end to end: from a raw MySQL database, through cleaning and verification, into exploratory analysis, charts, and written business insights covering 915 employees across eight departments.

## OBJECTIVE

- Understand overall attrition and which departments are most at risk
- Examine whether pay, tenure, and overtime differ between active and exited employees
- Test whether job satisfaction actually predicts performance
- Identify the highest paying roles and largest departments
- Translate patterns into concrete retention recommendations

## TOOLS & ENVIRONMENT

### Python 3

Core programming language

### Pandas

Cleaning, transformation, aggregation

### NumPy

Numerical insight calculations

### Matplotlib & Seaborn

Chart generation

### SQLAlchemy & PyMySQL

Database connectivity

### MySQL

Source relational database, 8 tables

# Dataset Overview

*Eight linked MySQL tables, verified and joined into one 915 row master table*

| Table               | Raw Rows | Cleaned Rows | Purpose                                      |
|---------------------|----------|--------------|----------------------------------------------|
| locations           | 12       | 12           | City, state, country reference data          |
| departments         | 10       | 8            | Department names; 2 duplicate entries merged |
| jobs                | 30       | 30           | Job titles with defined salary bands         |
| employees           | 915      | 915          | Core employee records                        |
| job_history         | 1,307    | 1,307        | Role and department change history           |
| performance_reviews | 1,848    | 1,848        | Rating, satisfaction, work-life balance      |
| attendance          | 8,400    | 8,400        | Daily status and overtime hours              |
| attrition           | 146      | 146          | Exit date and reason                         |

*Every one of the 915 employee records survived cleaning. Departments dropped from 10 to 8 because two entries, "HR" and "Ops", were the same real department entered twice under different IDs.*

# Data Pipeline Architecture

*Five sequential stages, each depending on the output of the one before it*



Each script processes one table at a time, called manually in sequence, so a failure in one table cannot silently corrupt the rest of the pipeline. The master table is only built once all eight tables pass verification.

# Data Cleaning: Key Decisions

*Every fix was evidence based: correct what is clearly wrong, avoid deleting records where possible*

## Merged duplicate departments

"HR" and "Ops" existed under two different IDs. Remapped and merged into full department names.

## Corrected out of range salaries

Salaries below a job's minimum were raised to it; salaries above the maximum were capped, with every correction counted.

## Fixed negative distance values

EDA found a minimum of -52 km. Corrected using absolute value instead of dropping the row.

## Standardized text formatting

Fixed inconsistent casing across cities, departments, and job titles, then repaired acronyms like IT and HR broken by title-casing.

## Filled missing values with clear logic

Missing salary filled by job role median; missing exit reason labeled "Not Specified" rather than left blank.

## Clipped invalid overtime hours

Negative overtime values, which cannot exist, were clipped to zero.

# Exploratory Data Analysis

*Key workforce metrics across all 915 employees*

**915**

TOTAL EMPLOYEES

**769**

ACTIVE (84.0%)

**146**

EXITED (16.0%)

**61,266**

AVERAGE SALARY

**8.13 yrs**

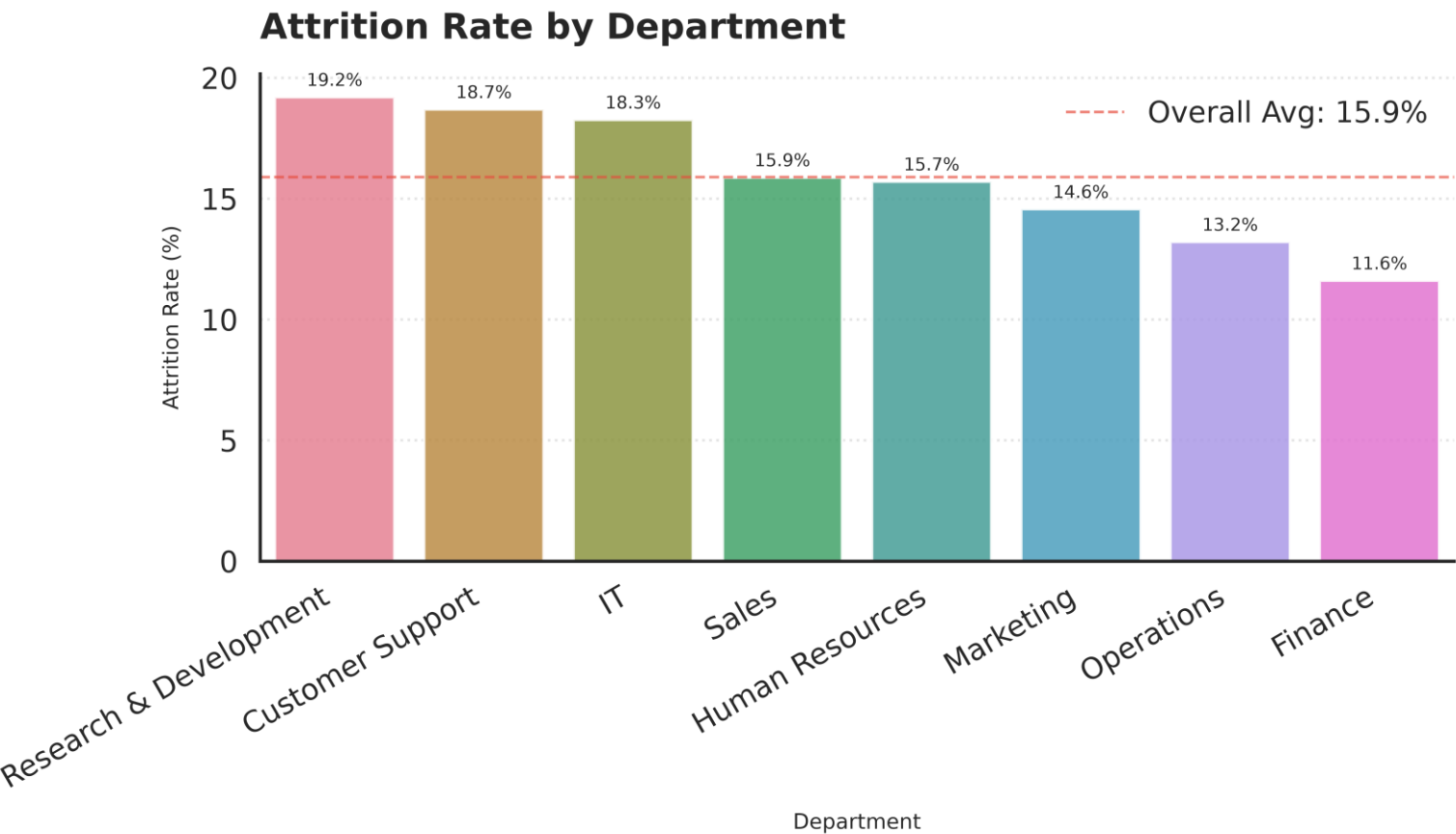
AVERAGE TENURE

**6.18 yrs**

AVG TENURE, EXITED

Gender: 491 Female (53.7%) | 424 Male (46.3%)    Departments: 8, ranging 103 to 126 employees each

# Attrition Rate by Department

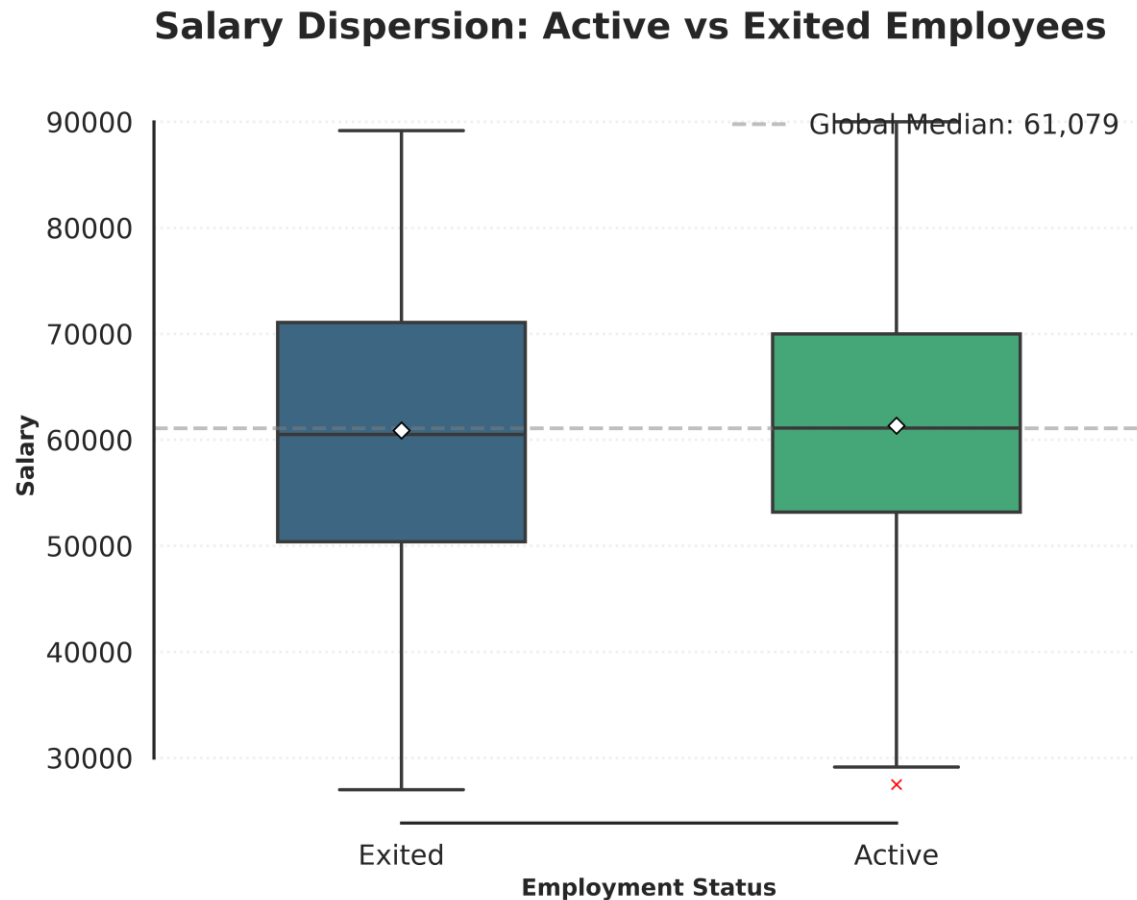


KEY TAKEAWAYS

- R&D runs hottest:** Research & Development has the highest attrition at 19.20%, followed by Customer Support (18.69%) and IT (18.26%).
- Finance is most stable:** Lowest attrition at 11.61%, with Operations close behind at 13.21%.
- The gap is real:** Roughly 7.6 percentage points separate the highest and lowest department attrition rates.



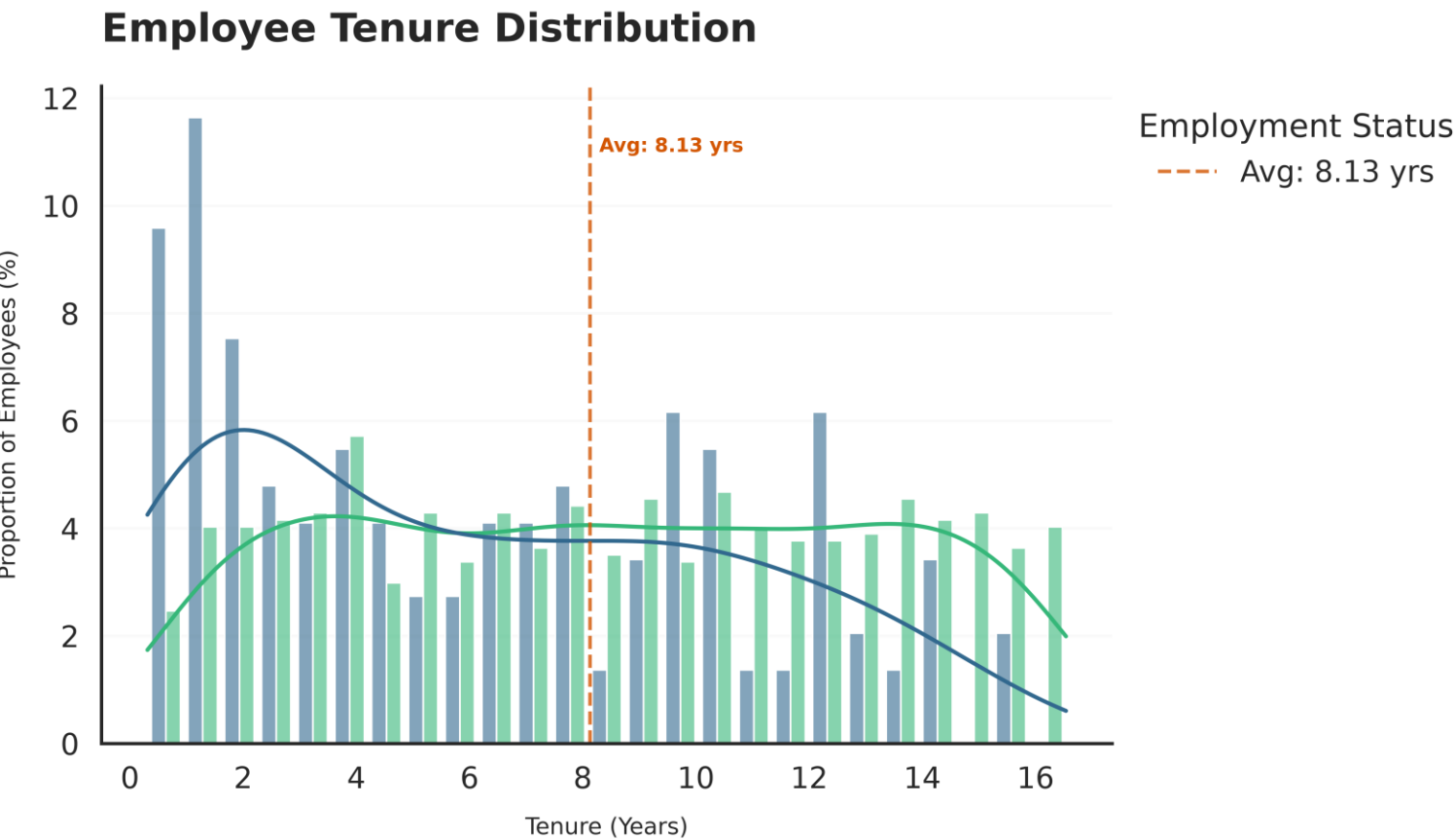
## Salary Dispersion: Active vs Exited



### KEY TAKEAWAYS

- **Pay is not the driver:** Active employees earn just 452 more on average than exited employees, a gap under 1%.
- **Medians agree:** 61,116 median for active vs 60,506 for exited, a very small gap given the full salary range of 27,000 to 90,000.
- **Implication:** Compensation alone is unlikely to explain attrition here; look at role and workload factors instead.

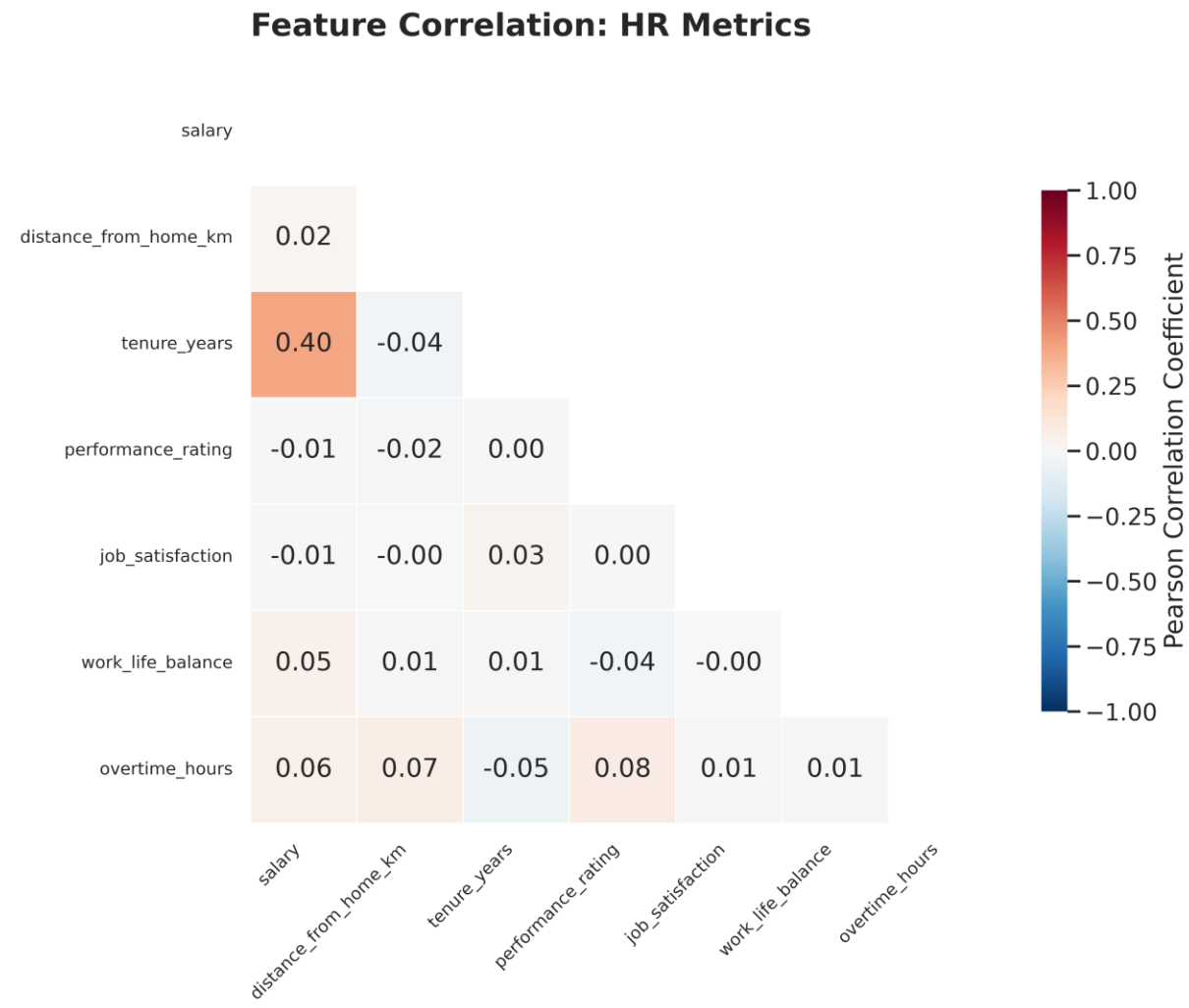
# Employee Tenure Distribution



## KEY TAKEAWAYS

- **Exits happen earlier:** Employees who left stayed 6.18 years on average, short of the company-wide average of 8.13 years.
- **Risk is front loaded:** Attrition risk concentrates in the earlier and mid stages of tenure, not spread evenly.
- **Retention window:** The first six to eight years appear to be the critical window to focus retention efforts.

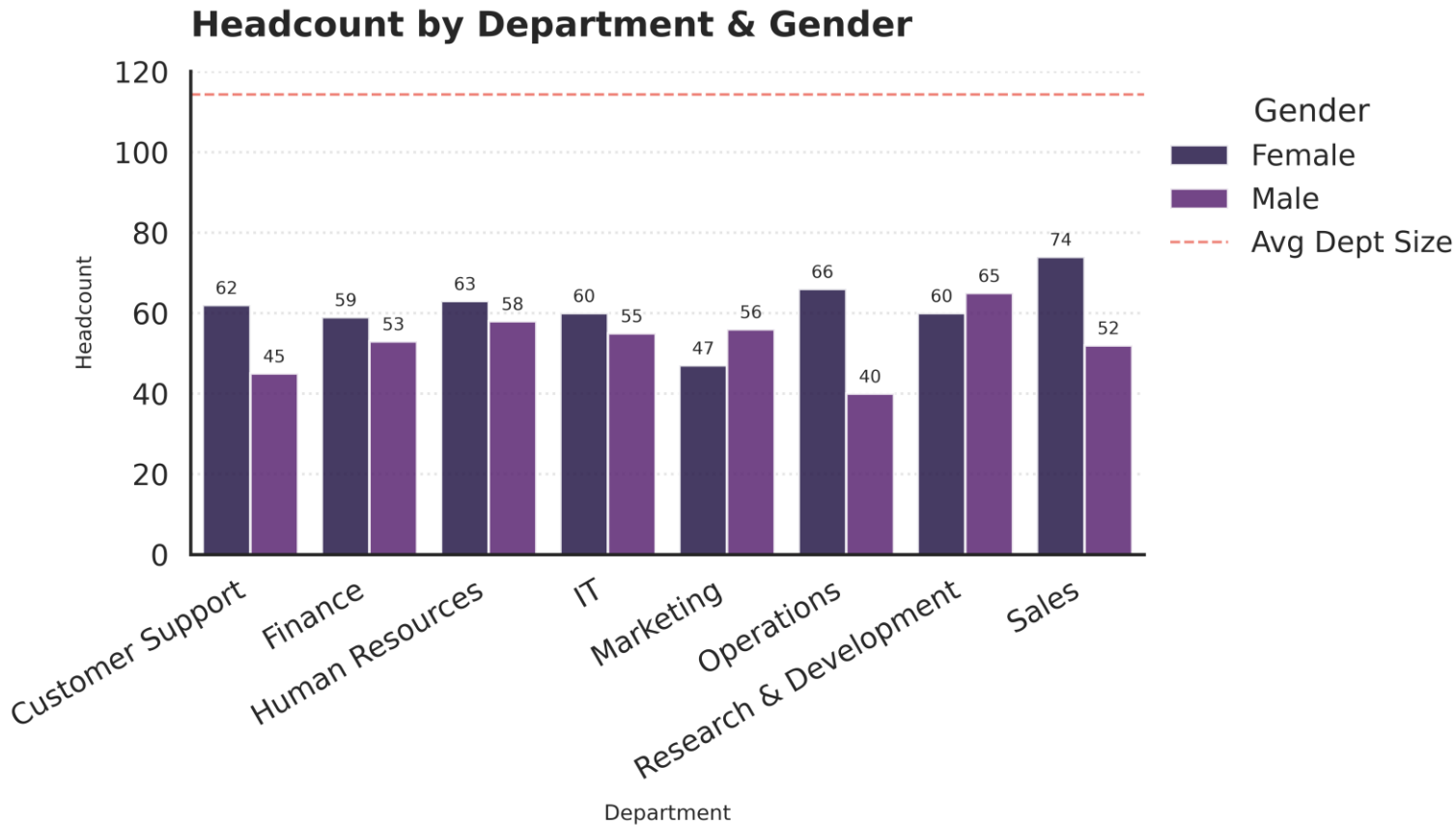
# Feature Correlation Heatmap



KEY TAKEAWAYS

- **Tenure tracks salary:** Salary and tenure correlate at 0.40, the strongest relationship on the board, consistent with standard pay progression.
- **Satisfaction does not predict performance:** Job satisfaction and performance rating correlate at essentially 0.00.
- **Everything else is flat:** Overtime, distance from home, and work life balance all show weak correlations under 0.10 with pay and performance.

## Headcount by Department & Gender



### KEY TAKEAWAYS

- **Departments are close in size:** All eight fall in a narrow band of roughly 103 to 126 employees each.
- **Gender balance holds company-wide:** 53.7% female and 46.3% male overall, reasonably consistent across departments.
- **A clean baseline:** Because department sizes are close to even, the attrition comparisons earlier are not distorted by size.

# HR Insights: Questions 1 to 4

*Answered directly from the 915 row dataset using NumPy*

## Q1. What is the overall attrition rate?

15.96% of employees have left the company, 146 out of 915.

## Q2. Which department has the highest attrition rate?

Research & Development, at 19.20%, the highest of all eight departments.

## Q3. Is there a salary gap between active and exited employees?

Barely. Active employees earn 452 more on average (mean 61,338 vs 60,886 for exited).

## Q4. What is the average employee tenure?

8.13 years overall. Employees who left stayed 6.18 years on average before exiting.

# HR Insights: Questions 5 to 8

*Answered directly from the 915 row dataset using NumPy*

## **Q5. Is attrition different between genders?**

Yes, modestly. Male employees leave more often at 16.75% vs 15.27% for female employees.

## **Q6. Are happier employees also better performers?**

No measurable link. Job satisfaction and performance rating correlate at 0.00.

## **Q7. Which job title earns the most on average?**

IT Manager, at 78,599, just ahead of Software Engineer (78,054) and System Admin (77,652).

## **Q8. Do employees who work more overtime leave more often?**

Slightly. Exited employees averaged 1.15 overtime hours vs 1.08 for active employees, suggesting overwork may play a role.

# Real-World Problems & Solutions

1

## R&D, Support, and IT are bleeding talent

### THE DATA

R&D (19.20%), Customer Support (18.69%), and IT (18.26%) all sit well above the 15.96% company average.

### THE SOLUTION

Since the salary gap is negligible, target exit interviews and workload reviews in these three departments rather than a pay based fix.

2

## Attrition risk concentrates early

### THE DATA

Employees who leave do so after 6.18 years on average, well short of the 8.13 year company average.

### THE SOLUTION

Focus retention programs and career check-ins on the three to seven year tenure band.

3

## Overtime load tracks with who leaves

### THE DATA

Exited employees logged 1.15 average overtime hours vs 1.08 for active employees.

### THE SOLUTION

Pair overtime tracking with performance data to flag rising burnout risk in strong performers before they resign.

# Strategic Recommendations

1

## **Prioritize R&D, Support, and IT**

These three departments run 2 to 3 points above the company attrition average; start exit interviews here first.

2

## **Do not lead with pay**

The active vs exited salary gap is under 1%. Investigate workload and management quality instead of compensation.

3

## **Build a tenure based retention program**

Concentrate development conversations on the three to seven year tenure band, where exits are concentrated.

4

## **Track satisfaction and performance separately**

With a 0.00 correlation between them, do not treat one as a proxy for the other in reviews.

5

## **Watch overtime in high performers**

Exited employees carried a higher average overtime load; pair overtime and performance data to catch burnout early.



# Conclusion

This pipeline takes 915 employee records through a full cleaning, verification, and analysis workflow, from a raw eight-table MySQL schema to five recruiter-ready charts and eight quantified business insights. Attrition here is not primarily a pay problem: department, tenure stage, and overtime load show a far stronger relationship with who leaves than compensation does.

## DATA DISCLAIMER & SCOPE

This report is based on a sample HR dataset of 915 employee records built for a data analytics portfolio project, not a live production system. Some fields have partial coverage: overtime\_hours is present for 400 of 915 employees, and performance\_rating is missing for 83. Every statistic using these fields draws only on employees with a valid value. Findings here should be read as a demonstration of method and a directional read on this dataset, not a substitute for a live HR system's ongoing monitoring.